

Content Based Filtering Report

1. Description of Feature Extraction & Cleaning

For building the content-based filtering system, the following metadata fields were selected from the dataset:

- Title
- Author
- Publisher

Data preprocessing steps included:

- Converting all text fields to lowercase to avoid case mismatch during similarity computation.
- Removing special characters and excess spaces to clean the textual data.
- Handling missing values by filling them with placeholder text ("Unknown") where necessary.
- Combining the selected features into a single "metadata soup", effectively merging the title, author, and publisher into a unified text representation for each book.

This ensured that the vectorizer could capture meaningful signals across all provided metadata.

2. Vectorization Technique and Why It Was Chosen

Technique Used:

TF-IDF Vectorization (Term Frequency-Inverse Document Frequency)

Why TF-IDF?

- TF-IDF reduces the influence of common words (like "the", "a", etc.) and highlights more meaningful, distinguishing terms (like "hobbit", "tolkien").
- It is more effective than simple counting because it assigns weights based on term importance.

- It works very well for sparse and high-dimensional text data, which is typical when working with book metadata.

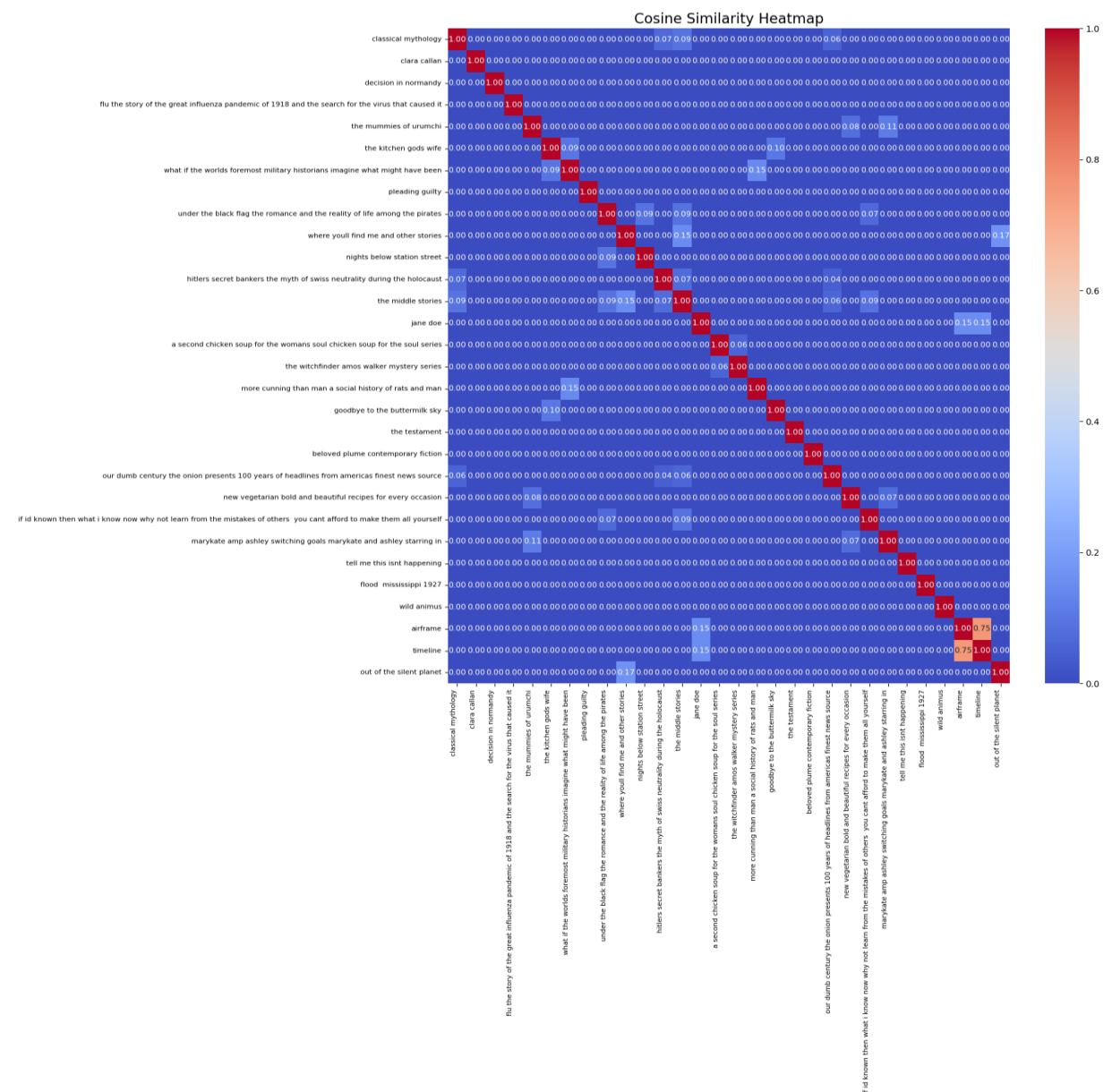
The TF-IDF vectorizer created a feature matrix where:

- Rows represent books
- Columns represent unique terms across all metadata

This matrix served as the basis for computing book-to-book similarity.

3. Screenshots of Similarity Matrix & Output Examples

A heatmap was generated based on the cosine similarity matrix for a subset of 30 books.



- Brighter colors indicate higher similarity.
- Darker colors indicate lower similarity.

Book: classical mythology
 Top 3 similar books: the middle stories, hitlers secret bankers the myth of swiss neutrality during the holocaust, our dumb century the onion presents 100 years of headlines from americas finest news source

Book: clara callan
 Top 3 similar books: out of the silent planet, timeline, decision in normandy

Book: decision in normandy
 Top 3 similar books: out of the silent planet, timeline, clara callan

Book: flu the story of the great influenza pandemic of 1918 and the search for the virus that caused it
 Top 3 similar books: out of the silent planet, timeline, clara callan

Book: the mummies of urumchi
 Top 3 similar books: marykate amp ashley switching goals marykate and ashley starring in, new vegetarian bold and beautiful recipes for every occasion, out of the silent planet

Book: the kitchen gods wife
 Top 3 similar books: goodbye to the buttermilk sky, what if the worlds foremost military historians imagine what might have been, out of the silent planet

Book: what if the worlds foremost military historians imagine what might have been
 Top 3 similar books: more cunning than man a social history of rats and man, the kitchen gods wife, out of the silent planet

Book: pleading guilty
 Top 3 similar books: out of the silent planet, timeline, clara callan

Book: under the black flag the romance and the reality of life among the pirates
 Top 3 similar books: nights below station street, the middle stories, if id known then what i know now why not learn from the mistakes of others you cant afford to make them all yourself

Book: where youll find me and other stories
 Top 3 similar books: out of the silent planet, the middle stories, timeline

Book: nights below station street
 Top 3 similar books: under the black flag the romance and the reality of life among the pirates, out of the silent planet, timeline

Observations:

- "Out of the Silent Planet" appears very frequently as a recommendation, suggesting it's centrally similar to many items based on the metadata soup.

4. Evaluation

Mean Cosine Similarity of Top Recommendations

- The average cosine similarity between each book and its top 5 recommended books was calculated.
- Results:
 - Classical Mythology → 0.043
 - The Kitchen God's Wife → 0.0395
 - Clara Callan → 0.000

These low scores suggest that even the most similar recommended books were only weakly related based on the available metadata (title, author, publisher).

5. Insights & Observations from Recommendations

- Dominance of Certain Books:
 - "*Out of the Silent Planet*" appears frequently in the top recommendations for many different queries.
 - Suggests that the book's metadata (title, author, publisher) is broadly similar to many others, possibly due to generic features captured by TF-IDF.
- Mixed Thematic Relevance:
 - Some recommended books are not thematically aligned.
 - Example: "*Flu*" (about a pandemic) recommending "*Out of the Silent Planet*" (sci-fi novel).
 - Shows that metadata fields alone (without genre or content summary) are not always enough for true thematic matching.
- Publisher and Title Influence:
 - Books with similar publishers or similar title wording were often grouped together.
 - Even when the actual content or genre differed significantly.
- Metadata Limitations:
 - The model primarily relies on surface-level similarities.
 - Without fields like genres, summaries, or detailed descriptions, deeper content-based matching is limited.
- Potential Improvements:
 - Adding more metadata fields such as:
 - Genre categories
 - Book descriptions
 - Keywords or tags
 - This would help improve the recommendation quality and ensure more meaningful matches.

6. Comparative Notes vs Collaborative Filtering

Aspect	Content-Based Filtering	Collaborative Filtering
Data Used	Book metadata (e.g., title, author, publisher)	Rating dataset (e.g., book_id ,user_id , ratings)
Strength	Works even for new users and new books (no cold start problem)	Captures hidden user patterns; can suggest unexpected favorites
Weakness	Can't easily suggest completely different books	Suffers from cold start for new users or new items
Explainability	High — similarity reasons (title, author) are transparent	Low — often hard to explain why something is recommended
Dependency	Relies on metadata quality	Relies on large-scale user interaction data
Drawbacks	Surface-level similarity; limited creativity	Needs large active userbase; cold start issues

In summary, while content-based filtering offers simplicity and explainability, it struggles to suggest diverse or unexpected books compared to collaborative filtering, which relies heavily on rich user interaction data.